

SalonOS

Inventory, Financial Flow & Reporting Specification

Source-of-truth flows, report lineage, formulas, tax treatment, COGS, refunds, supplier ledger, payment accounts and report-page blueprint

Prepared for	Aliyan Salon Management / SalonOS
Prepared by	iSysware project workflow
Status	Locked functional/reporting blueprint before implementation

Critical accounting lock

Tax is a liability, not income. Tips are staff liability, not income. Staff commission is calculated only on attributed net sales after discounts and valid reversals/refunds, excluding tax and tips. Inventory purchases and supplier payments are not COGS until stock is sold or consumed.

Version 1.0 - 01 October 2026

1. Purpose and Locked Reporting Scope

This specification defines exactly where each reporting figure comes from, how inventory and financial events flow through SalonOS, and how calculations must behave under normal sales, partial payments, refunds, voids, stock consumption, expiry, supplier credit and account movements. The purpose is to prevent double counting and keep every report reconcilable to canonical transactions.

Final Reporting & Analytics menu - only these 11 pages

#	Report	Purpose
1	Income & Expense	Recognized business income and recognized operating expenses; excludes tax, tips, transfers and supplier-payable settlement.
2	Sales & Invoices	Posted invoices, discounts, tax, tips, payments, outstanding, void/refund lifecycle and sales reversals.
3	Payment Accounts	Named bank/online accounts: opening, money in, money out, transfers and closing balance.
4	Cash Drawer Logs	Physical cash custody by user/session, expected cash, counted cash, variance and settlements.
5	Detailed Expenses	Expense transactions by category, user, branch and payment source.
6	Staff Salary Report	Finalized payroll, earnings/deductions, payment and outstanding salary.
7	Staff Commission Report	Attributed net sales, rate snapshots, commission earned/reversed/paid/outstanding.
8	Staff Performance	Completed work, attributed sales, visits, attendance and performance metrics.
9	Inventory Reports	Stock, valuation, purchases, movements, suppliers, expiry, consumption, COGS and product profit.
10	Attendance & Overtime	Attendance, leave, hours, lateness and authorized manual overtime.
11	Operating Profit	Net sales less product COGS, material consumption and recognized operating expenses.

No extra report-menu pages

Tax, appointment, tips statement, unpaid invoices and settlement history may still exist as source-module views or subviews, but they are not separate items in the final Reporting & Analytics sidebar. Tax summaries live inside Sales & Invoices / Payment Accounts / Operating Profit supporting totals.

Universal reporting controls

- Branch filter. Super Admin/Owner can use All Branches or one authorized branch; Admin/Accountant remain branch-scoped.
- Date presets: Today, Yesterday, This Week, This Month, Last Month and Custom Range. Every page must state what its date filter means.
- Owner/Super Admin can filter any authorized report by User Type and specific User where the report has an actor dimension.
- Exports: filtered PDF and filtered Excel. Print uses the same filtered report content. Export metadata includes branch, date range, filters, generated timestamp and generated-by user.
- Reports are read-only. Operational mutations remain in source modules. Sales & Invoices may provide a permission-aware View/Open Invoice handoff for void/refund actions.
- All totals derive from canonical source records; report code must not independently invent balances.

2. End-to-End Source-of-Truth Flow

Golden rule

A report never becomes the source of truth. Operational modules post immutable or auditable events; reports aggregate those events using explicit date and status rules.

Source event	Canonical record	Report consequence
Appointment	Customer + schedule + quote + assignments	No sale, no commission, no stock movement until POS posting.
POS Invoice	Final performed lines, discounts, tax, tips, staff attribution	Creates sale history; can link appointment; invoice history remains immutable.
Payment Receipt	Cash / named online account / split payment	Moves money into payment account or cash drawer; does not create a second sale.
Refund / Void / Credit	Dated reversal/credit event	Reduces effective sale; may refund money; never deletes original invoice.
Purchase	Supplier + inventory cost layers + payment terms	Increases inventory; credit portion increases supplier payable.
Supplier Payment	Supplier ledger + cash/online outflow	Reduces payable/advance; never creates COGS.
Stock Movement	Purchase in, POS sale out, consumption, return, adjustment	Only mechanism that changes quantity on hand.
Payroll Finalization	Salary expense/liability snapshot	Salary payment settles liability; must not expense salary twice.
Commission Event	Earn/reversal liability from attributed net sales	Payment settles commission liability; does not create second expense.
Tip Event	Tip collection/allocation/payout/reversal liability	Tip never enters revenue or commission base.
Expense	Recognized operating expense + payment source	Affects expenses and, if paid, cash/online account.
Settlement/Transfer	Custody/account transfer	Moves money between holders/accounts; not income or expense.

Data lineage by report

Report	Primary sources	Default date basis
Income & Expense	Invoices/refunds, expenses, finalized payroll, commission earn/reversal, inventory write-offs if classified as operating loss	Recognition date; not payment date unless the source event itself is cash-basis.
Sales & Invoices	Invoices, invoice lines, receipts, credit/refund events, linked customer/staff snapshots	Invoice posting date for sales; refund event date for reversal.
Payment Accounts	Account movement ledger	Movement timestamp.
Cash Drawer Logs	Drawer sessions + cash movements + settlements	Drawer movement/session timestamp.
Detailed Expenses	Expense records + payment account/drawer link	Expense date/posting date.
Salary	Finalized payroll + salary payment records	Payroll period / finalized date; payment date separately.
Commission	Commission earn/reversal events + payouts	Commission event date; payout date separately.
Staff Performance	Posted invoice attribution + completed appointments + attendance	Underlying performance event date.
Inventory	Stock movements, batches, purchases, supplier ledger	Movement date; current valuation uses as-of date.
Attendance & Overtime	Attendance, leave, manual overtime	Work date.
Operating Profit	Sales/refunds + COGS + consumption + expenses + payroll + commission + write-offs	Recognition dates across source events.

3. Core Sales, Invoice, Tax and Payment Formulas

3.1 Invoice line and invoice totals

Line Gross = Quantity x Unit Selling Price

Line Net Sales = Line Gross - Line Discount - Allocated Invoice Discount

Invoice Net Sales = Sum(valid line net sales) - Valid Net-Sales Reversals

Tax Charged = Sum(Taxable Line Net Sales x Applicable Tax Rate)

Invoice Total = Net Sales + Tax + Tip

Tax and tip exclusion

Tax is collected on behalf of the tax authority and is a liability. Tip is collected on behalf of staff and is a liability. Neither is sales income, operating profit, COGS or staff commission base.

3.2 Payment and outstanding

Effective Invoice Total = Original Invoice Total - Approved Credit / Refund Document Total

Net Applied Payments = Valid Receipts - Customer Cash/Online Refunds that de-apply those receipts

Outstanding = max(0, Effective Invoice Total - Net Applied Payments)

Payment Status = UNPAID if Net Applied Payments = 0; PARTIALLY_PAID if 0 < Net Applied Payments < Effective Invoice Total; PAID if Net Applied Payments >= Effective Invoice Total

- Invoice lifecycle is independent: ACTIVE, PARTIALLY_REFUNDED, REFUNDED or VOIDED can coexist with the payment-state logic.
- A BILLED appointment can still have an UNPAID or PARTIALLY_PAID invoice.

3.3 Tax treatment

Net Tax Liability = Tax Charged on valid posted invoices - Tax Reversed on approved refund/credit events

Taxable Base = Net taxable line amount after discounts; tax is never calculated on the pre-discount amount unless tax policy explicitly requires it.

Tax Refund/Reversal = Refunded Taxable Net Amount x Original Tax Rate Snapshot

Tax collection display

Sales & Invoices should show Tax Charged, Tax Reversed and Net Tax Liability. For partially paid invoices, do not call all invoiced tax "cash collected". If a cash-realized tax KPI is required, derive it from explicit receipt allocation policy; otherwise keep invoice-basis tax and cash receipts separate.

3.4 Example - normal taxable sale

Component	Calculation	Amount
Gross service	PKR 10,000	10,000
Discount	PKR 1,000	-1,000
Net sales	10,000 - 1,000	9,000

Component	Calculation	Amount
Tax @ 10%	9,000 x 10%	900
Tip	Separate liability	500
Invoice total	9,000 + 900 + 500	10,400
Revenue used in reports	Net sales only	9,000
Commission base	Attributed part of net sales only	Excludes 900 tax and 500 tip

4. Staff Attribution and Commission Formulas

Commission must be computed from the final posted sale, not from appointment confirmation. The final invoice stores historical staff assignments, net allocation amounts and rate snapshots so later catalogue or staff-rate edits cannot rewrite history.

4.1 Individual service

Service Attributed Net Sales = Service Line Net Sales after all applicable discounts

Staff Commission Earned = Service Attributed Net Sales x Commission Rate Snapshot

4.2 Package with multiple staff

Package Net Sales = Package Selling Price - Package/Invoice Discounts Allocated to Package

Component Net Allocation = Package Net Sales x Saved Component Allocation Weight

Staff Attributed Net Sales = Sum(Component Net Allocation for components performed by that staff member)

Staff Commission = Staff Attributed Net Sales x Applicable Commission Rate Snapshot

Locked rule

Adding more staff to a package never multiplies the package price. Staff share is an attribution split of the single package net sale.

Example	Ali	Ahmed
Package net sale	PKR 5,000 total	PKR 5,000 total
Weights performed	40% + 20%	40%
Attributed net sales	PKR 3,000	PKR 2,000
Rate	10%	15%
Commission	PKR 300	PKR 300

4.3 Discount, refund and reversal handling

Commission Base = Attributed Net Sales AFTER discount, EXCLUDING tax and tip

Commission Reversal = Refunded / Reversed Attributed Net Sales x Original Commission Rate Snapshot

Net Commission Liability = Commission Earned - Commission Reversed - Valid Commission Payouts

- A refund dated in October creates an October commission reversal; it must not erase September earned commission.
- A commission payout reduces commission payable/custody; it is not a second commission expense if the earn event already recognized the expense.

- If only one package component is refunded, reverse only that component attribution (plus its allocated discount basis), not the whole package unless the whole package is reversed.

4.4 Tips

Tip Liability Closing = Opening Tip Liability + Tips Collected + Reversals Restoring Liability - Tip Payouts - Valid Tip Refunds/Reductions

Cross-date rule

If a tip payout occurs on Sep 30 and is reversed on Oct 1, September keeps the payout and October records the dated reversal restoring liability. Current REVERSED status must never erase the historical Sep 30 event.

5. Inventory Costing, Valuation and Stock Flow

Inventory is branch-scoped and movement-driven. Current quantity is never directly edited. Every change is a posted stock movement with a source reference, batch where applicable, cost snapshot, actor and timestamp.

5.1 Purchase landed cost

Line Gross Purchase Cost = Purchase Quantity x Entered Unit Purchase Price

Line Net Merchandise Cost = Line Gross Purchase Cost - Line Discount

Allocated Header Discount = Header Discount x (Line Net Merchandise Cost / Sum All Line Net Merchandise Costs)

Allocated Capitalizable Charges = Capitalizable Charges x (Line Net Merchandise Cost / Sum All Line Net Merchandise Costs)

Landed Line Cost = Line Net Merchandise Cost - Allocated Header Discount + Allocated Capitalizable Charges

Landed Unit Cost = Landed Line Cost / Received Quantity

- Non-capitalizable ancillary expenses are posted to Detailed Expenses, not buried inside inventory cost.
- Historical landed unit cost is immutable after purchase posting.

5.2 Quantity and current valuation

Batch Quantity On Hand = Sum(Posted Quantity In) - Sum(Posted Quantity Out)

Item Quantity On Hand = Sum(All Batch / Cost-Layer Quantities On Hand for branch)

Inventory Cost Value = Sum(Remaining Quantity x Historical Landed Unit Cost)

Potential Retail Value (informational only) = Sellable Quantity x Current POS Selling Price

Do not confuse values

Inventory Cost Value is an asset/cost figure. Potential Retail Value is only an informational selling-value estimate and must never be used as COGS or operating profit.

5.3 Cost-layer selection

- Expiry-tracked stock: use FEFO (First Expire, First Out) among valid, non-quarantined batches.

- Non-expiry stock: use FIFO by received date unless an existing item-specific costing policy is already configured.
- Expired or quarantined stock is blocked from normal POS sale and normal salon consumption.
- Adjacent/partial quantities may consume multiple cost layers; COGS is the exact sum of layer quantities x layer costs.

5.4 Product COGS and in-house consumption

Product Sale COGS = Sum(Sold Quantity from each cost layer x that layer Landed Unit Cost)

Product Gross Profit = Product Net Sales - Product Sale COGS

In-House / Salon Consumption Cost = Sum(Consumed Quantity x cost-layer unit cost)

Inventory Write-off Cost = Sum(Expired/Damaged/Missing Quantity x cost-layer unit cost)

6. Inventory and Supplier Scenario Rules

Scenario	Postings	Reporting effect
Credit purchase	Inventory +100,000; Supplier Payable +100,000; Cash 0	No COGS at purchase time.
Cash purchase	Inventory +100,000; Cash/Account -100,000; Supplier payable 0	Still no COGS until sold/consumed.
Partial purchase	Inventory +100,000; Cash -40,000; Supplier Payable +60,000	Paid and payable portions stay separate.
Supplier payment later	Supplier Payable -60,000; Cash/Bank -60,000	Not expense and not COGS.
Supplier return	Inventory decreases at original layer cost; payable/credit reduces	Original purchase remains in history.
POS retail sale	Stock OUT; COGS at selected layer cost; sales revenue at net selling amount	Tax/tip excluded from revenue and COGS.
In-house salon use	Stock OUT; material-consumption cost recognized	No sales revenue.
Expired/damaged	Stock OUT; inventory write-off/loss recognized	Not normal product COGS.
Physical shortage	Settlement creates negative adjustment at approved cost basis	Requires reason and actor.
Physical surplus	Settlement creates positive adjustment at approved cost basis	Requires approved cost source.
Branch transfer	Source OUT + destination IN at same historical cost	No revenue, COGS or profit.
Customer return - resalable	Refund/credit reverses sales; RETURN_IN restores stock at original sale cost; COGS reversal	Original invoice remains.
Customer return - damaged	Return can be recorded into quarantine then DAMAGED_OUT at original cost	Revenue reverses; cost becomes inventory loss/write-off.
Goodwill refund, product not returned	Revenue may reverse per approved refund; stock and original COGS stay	Do not fake stock return.

6.1 Supplier ledger

Supplier Closing Payable = Opening Payable + Credit Purchases - Supplier Payments - Purchase Returns/Credits +/- Approved Adjustments

If result < 0, display absolute amount as Supplier Advance/Credit; do not clamp to zero.

6.2 Inventory report KPIs

KPI	Formula / source
Current Inventory Cost Value	Sum remaining stock layers x landed cost
Purchased Stock Value	Posted purchase landed cost in selected period
Retail Product Net Sales	Posted retail product line net sales less valid product-sale reversals

KPI	Formula / source
Product COGS	Cost of posted sold quantities less valid COGS reversals
Product Gross Profit	Retail Product Net Sales - Product COGS
In-house Used Quantity	Sum salon/internal consumption OUT quantities
In-house Use Cost	Sum consumption quantities x historical layer cost
Expired/Damaged Quantity	Sum explicit expiry/damage OUT quantities
Write-off Cost	Expiry/damage/missing quantity x layer cost
Supplier Payable	Running supplier ledger closing payable
Near Expiry	Valid batch qty with expiry within configured alert window
Out of Stock	Active branch items with usable qty <= 0

7. Payment Accounts, Bank/Online Accounts and Cash Drawers

Payment Accounts report is the mandatory bank/online reconciliation view. It answers: how much was in the account at the start, how much came in, how much went out, what transfers occurred, and what remains. Physical cash is controlled separately through user cash drawers and settlements.

7.1 Named payment-account formula

Account Closing Balance = Opening Balance + Valid Money In - Valid Money Out + Transfer In - Transfer Out +/- Approved Adjustments

Money In examples	Money Out examples	Not income/expense by itself
POS/dues online receipt	Customer refund	Account transfer in/out
Supplier cash/online refund	Supplier payment	Cash-drawer settlement transfer
Settlement received into account	Operating expense	Opening balance migration
Other approved receipt	Salary/commission payout	Approved correction/transfer

Important distinction

A payment account can receive PKR 10,400 for a sale whose business revenue is only PKR 9,000 because the bank receipt also contains PKR 900 tax and PKR 500 tip liability. Account cash flow and income are intentionally different.

7.2 Cash drawer formula

Expected Physical Cash = Opening Float + Cash Sales/Receipts + Approved Cash In - Cash Refunds - Cash Expenses - Accepted Settlements Out +/- Approved Cash Adjustments

Drawer Variance = Physical Count - Expected Physical Cash

Remaining Drawer Cash = Expected Physical Cash - Accepted Settlement Amount (or actual configured carry-forward rule)

- Drawer opening float is custody, not income.
- Cash settlement to Admin/Super Admin is custody transfer, not income or expense.
- Old dues collected in cash increase drawer cash but do not create a new sale.
- Tip cash received increases drawer cash but the tip amount remains staff liability, not business income.

7.3 Account report required columns

Date/Time	Account	Type	Reference	User	Money In	Money Out	Running Balance	Source
2026-10-01 10:00	HBL Smart POS	Invoice receipt	INV-...	Cashier	10,400	-	...	POS
2026-10-01 12:00	HBL Smart POS	Supplier payment	SUPPAY-...	Admin	-	5,000	...	Supplier Ledger

8. Reporting UI/UX Blueprint

The final reporting redesign should follow the compact school-management reference: one-screen, line-oriented, dense but readable, with a single horizontal filter row where practical. Avoid oversized dashboard cards and operational controls inside reports.

Screen pattern

Title + one-line description -> compact filter row -> concise KPI/total strip -> dense report table -> footer totals -> PDF / Excel / Print. Mobile collapses filters into a filter sheet while preserving the same report fields.

8.1 Universal filter row

Control	Behavior
Branch	All authorized branches for Super Admin/Owner; fixed assigned branch for scoped roles.
Date Preset	Today, Yesterday, This Week, This Month, Last Month, Custom Range.
From / To	Enabled for Custom Range; clear timezone/date-basis label.
User Type	Where actor dimension applies.
User	Owner can select any authorized user; scoped roles only see authorized users.
Report-specific filters	Status, payment method, category, item, supplier, staff, invoice state, etc.
Apply / Reset	Explicit filter application if needed for performance; preserve filter state during drill-down.

8.2 Export rules

- PDF: salon name/logo, report title, branch, date basis/range, active filters, generated by, generated at, KPI totals and table.
- Excel: active filtered rows only, typed numeric/date columns, separate summary sheet/rows where useful, no screenshots.
- Print: same report semantics as PDF; no operational buttons printed.
- Exports must preserve PKR precision/rounding exactly as on-screen.

8.3 Monetary rounding

Store monetary calculations in integer smallest currency unit where possible; otherwise use decimal arithmetic, never binary floating-point for authoritative totals.

Component/package allocations: calculate high precision -> round final displayed currency -> assign any rounding remainder deterministically to the largest/last eligible component so allocations exactly reconcile to package net amount.

9. Report Page Specifications - Financial and Sales

9.1 Income & Expense

Date basis: Recognition date of the business event. Sales use invoice/refund event dates; expenses use expense date; payroll/commission use finalized earn/reversal dates.

Filters: Branch, preset/range, income/expense source, category, user type, user, payment method (informational), status.

KPIs / totals: Net Sales Income; Other Valid Income; Operating Expenses; Salary Expense; Commission Expense; Inventory Write-off/Loss; Net Before/After Selected Cost Categories.

Table fields: Date, Type, Source, Reference, Branch, User, Description, Recognized Income, Recognized Expense, Payment Method, Status.

Calculation notes: Exclude tax, tips, supplier payments, inventory purchase asset postings, settlements and account transfers. Old dues collection is not new income.

9.2 Sales & Invoices

Date basis: Invoice posting date for sales; refund/credit event date for reversals. Payment dates remain separate.

Filters: Branch, date, cashier/user, customer, invoice #, payment status, invoice lifecycle, payment method, service/package/product, tax status.

KPIs / totals: Gross Sales; Discounts; Net Sales; Tax Charged; Tax Reversed; Tips; Invoice Total; Paid; Refunded; Outstanding; Active/Refunded/Voided counts.

Table fields: Invoice #, Date/Time, Customer, Branch, Sold By, Lines Summary, Gross, Discount, Net Sales, Tax, Tip, Total, Paid, Refunded, Outstanding, Payment Status, Lifecycle, Action/View.

Calculation notes: Original invoice never disappears. UNPAID/PARTIALLY_PAID/PAID is separate from ACTIVE/PARTIALLY_REFUNDED/REFUNDED/VOIDED. Report drill-down can open operational invoice flow.

9.3 Payment Accounts

Date basis: Account movement timestamp.

Filters: Branch, date, account, transaction type, source module, user, reference.

KPIs / totals: Opening Balance; Money In; Money Out; Transfer In; Transfer Out; Closing Balance.

Table fields: Date/Time, Account, Type, Reference, Description, User, Money In, Money Out, Transfer, Running Balance, Source Module.

Calculation notes: Bank/account inflows may contain revenue + tax + tips together. This report is cash/account reconciliation, not P&L.

9.4 Cash Drawer Logs

Date basis: Drawer session/movement timestamp.

Filters: Branch, date, cash-handling user, drawer status/session, movement type, reference.

KPIs / totals: Opening Float; Cash Receipts; Cash Refunds; Cash Expenses; Settlements; Expected Cash; Physical Cash; Variance; Remaining Cash.

Table fields: Session, User, Opened/Closed, Opening, Receipt In, Refund Out, Expense Out, Settlement Out, Expected, Physical, Variance, Status.

Calculation notes: Custody transfers and opening float are not income/expense. Link every movement to actual actor and source reference.

10. Report Page Specifications - Expenses, Staff and Inventory

10.1 Detailed Expenses

Date basis: Expense date/posting date.

Filters: Branch, date, category, entered by user, payment method/account, status, keyword/reference.

KPIs / totals: Total Expenses; Cash Expenses; Online Expenses; Category totals.

Table fields: Expense #, Date, Category, Title/Description, Branch, Entered By, Payment Method, Account, Amount, Reference, Status.

Calculation notes: Do not duplicate purchases as expenses. Only separate ancillary/non-capitalizable costs belong here when correctly posted as expense.

10.2 Staff Salary Report

Date basis: Payroll period with finalized date; payment date shown separately.

Filters: Branch, payroll period/date, staff, designation/role, payment status, payment method.

KPIs / totals: Basic; Attendance Impact; Overtime; Allowances; Deductions; Gross; Net Salary; Paid; Outstanding.

Table fields: Period, Staff, Branch, Basic, Attendance, OT, Allowances, Deductions, Gross, Net, Paid, Outstanding, Status, Paid Date, Processed By.

Calculation notes: Use finalized payroll snapshots. Salary payout settles salary liability and must not expense the salary a second time.

10.3 Staff Commission Report

Date basis: Commission earn/reversal event date; payout date shown separately.

Filters: Branch, date, staff, service/package/product source, invoice, status, payout status.

KPIs / totals: Attributed Net Sales; Commission Earned; Commission Reversed; Net Commission; Paid; Outstanding.

Table fields: Date, Staff, Invoice, Customer, Service/Package Component, Attributed Net Sales, Rate Snapshot, Earned, Reversed, Net, Paid, Outstanding.

Calculation notes: Commission base excludes discounts, tax and tips. Refund reversals use original attribution and rate snapshots.

10.4 Staff Performance

Date basis: Underlying performance event date: completed appointment, posted invoice attribution and attendance work date.

Filters: Branch, date, staff, service/category, customer/source if applicable.

KPIs / totals: Completed Visits; Services Performed; Attributed Net Sales; Average Ticket; Commission Earned; Attendance Days/Hours; OT; Optional repeat-client metrics.

Table fields: Staff, Visits, Services, Attributed Net Sales, Product Sales if attributed, Average Ticket, Commission, Attendance, Hours, OT.

Calculation notes: Do not count appointment + linked invoice as two visits. Appointment confirmation alone creates no revenue or performance sales.

10.5 Inventory Reports

Date basis: Movement date for period reports; as-of date for current stock valuation.

Filters: Branch, report type, date, category, item, supplier, batch, movement type, item type, user, stock status.

KPIs / totals: Current Cost Value; Purchased Value; Net Product Sales; Product COGS; Gross Product Profit; In-house Use Qty/Cost; Expiry/Write-off; Supplier Payable; Low/Near Expiry counts.

Table fields: Dynamic by subreport: stock, batch, movement, purchase, supplier ledger, valuation, COGS, consumption, settlement and expiry columns.

Calculation notes: Current inventory is valued at remaining historical cost layers, not selling price. Supplier payment is not COGS. In-house use is material-consumption cost, not sales.

11. Report Page Specifications - Attendance and Profit

11.1 Attendance & Overtime

Date basis: Attendance/work date.

Filters: Branch, date, staff, attendance status, shift, leave type, overtime status.

KPIs / totals: Present Days; Absent; Leave; Late; Worked Hours; Authorized OT Hours; OT Amount.

Table fields: Date, Staff, Shift, Check In, Check Out, Worked Hours, Late, Early Leave, Attendance Status, Leave, OT Hours, OT Rate, OT Amount, Entered/Approved By.

Calculation notes: Overtime remains manual/authorized under existing rules; biometric time alone must not silently create payable overtime.

11.2 Operating Profit

Date basis: Recognition date across sale/refund, COGS/consumption movement, expense, payroll, commission and write-off events.

Filters: Branch, date preset/range; optional department/category drilldown where supported. Owner can use All Branches.

KPIs / totals: Net Service/Package Sales; Net Product Sales; Total Net Sales; Product COGS; Salon Material Consumption; Gross Contribution; Operating Expenses; Salary; Commission; Inventory Write-off; Operating Profit; Margin %.

Table fields: Compact line-oriented P&L rows with current period, optional prior period comparison, amount and % of sales.

Calculation notes: Tax, tips, supplier payments, inventory purchases, dues collections, settlements and account transfers are excluded from profit. Refunds reduce sales on the dated reversal basis.

Total Net Sales = Net Service/Package Sales + Net Product Sales

Gross Contribution = Total Net Sales - Product COGS - Salon Material Consumption Cost

Operating Profit = Gross Contribution - Operating Expenses - Salary Expense - Commission Expense - Inventory Write-off/Loss - Other Recognized Operating Expenses

Operating Margin % = Operating Profit / Total Net Sales x 100 (when Total Net Sales > 0)

11.3 Why tax is not profit

Event	Bank/Cash movement	Revenue / Profit treatment
Customer pays invoice net 9,000 + tax 900 + tip 500	+10,400	Revenue +9,000 only; +900 tax liability; +500 tip liability
Tax remittance 900	-900	Reduces tax liability; not expense
Tip payout 500	-500	Reduces tip liability; not expense
Supplier payment 5,000	-5,000	Reduces supplier payable; not COGS/expense
Cash settlement between users 20,000	transfer only	No income/expense

12. Refund, Void and Reversal Scenarios

12.1 Fully paid invoice - full refund

Step	Amount	Effect
Original net sales	10,000	Recognized on original invoice date
Tax	1,000	Tax liability
Tip	500	Tip liability
Receipt	11,500	Account/drawer inflow
Full refund later	11,500	Customer cash/online outflow, dated refund
Sales reversal	10,000	Reduces net sales on refund date
Tax reversal	1,000	Reduces tax liability
Tip reversal/refund	500	Reduces/restores tip liability per actual tip state
Commission reversal	Based on 10,000 attributed net only	Never on tax/tip

12.2 Partially paid invoice - full void

Example

Original invoice total PKR 10,000; customer paid PKR 6,000; outstanding PKR 4,000. Full void creates a credit/reversal for the full valid invoice amount, cancels the remaining receivable, and refunds only the actual PKR 6,000 received if

policy requires cash return. Original invoice remains visible.

After full void: Effective Invoice Total = 0; Net Applied Payments = 0 after refund; Outstanding = 0

12.3 Partial refund

Remaining Net Sales = Original Net Sales - Approved Refunded Net Sales

Remaining Tax Liability = Original Tax - Refunded Tax

Remaining Commission = Original Commission - Commission Reversal on Refunded Attributed Net Sales

- Lifecycle becomes PARTIALLY_REFUNDED while payment status is recomputed against the reduced effective invoice total.

12.4 Product refund and stock

- Resalable product physically returned: create RETURN_IN at original sale cost and reverse corresponding COGS.
- Damaged/non-resalable return: return into quarantine then explicit damaged write-off; do not silently make it sellable.
- Goodwill refund without product return: reverse approved revenue/tax/commission basis as applicable, but do not create fake stock-in or COGS reversal.
- Retry/idempotency rules must prevent double refund, double stock return, double commission reversal or duplicate account movements.

13. Period Integrity, Opening/Closing Balances and Historical Events

Cross-date reporting must aggregate dated events, not infer history from the current status of a record. A current REVERSED, REFUNDED or VOIDED status must never erase what happened in a prior period.

13.1 Generic opening/closing formula

Opening Balance = Sum(all valid dated events before report start date)

Period Movement = Sum(all valid dated events inside report date range)

Closing Balance = Opening Balance + Period Movement

13.2 Applies to

- Payment account balances
- Supplier payable/advance
- Tip liability
- Commission liability
- Salary payable if tracked
- Inventory quantity/value by as-of date
- Customer receivables/outstanding dues
- Cash drawer custody/session balances where applicable

13.3 Historical snapshot rules

Snapshot	Must remain immutable after posting
Invoice	Item/service/package names, quantities, selling prices, discounts, tax rate/amount, staff assignments

Snapshot	Must remain immutable after posting
Package commission	Component instance IDs, allocation weights/net allocations, staff, rate snapshots
Inventory sale	Batch/cost-layer IDs and unit cost used for COGS
Purchase	Received quantities and landed cost layers
Payroll	Salary components/rates for finalized period
Refund/reversal	Original source reference and reversed amounts
Customer	Invoice-time customer snapshot even if profile later changes

14. Reconciliation Rules and Control Checks

Control	Required equality
Sales reconciliation	Sum invoice net sales - sales reversals = Sales & Invoices net sales = sales component used in Operating Profit.
Tax reconciliation	Tax charged - tax reversed = net tax liability; tax never appears in revenue/commission/profit.
Tip reconciliation	Tip collections +/- reversals - payouts = tip liability; tip never appears in revenue/commission.
Payment account reconciliation	Opening + in - out + transfers = closing for every named account.
Cash drawer reconciliation	Opening + cash in - cash out - accepted settlements +/- adjustments = expected cash; compare to physical count.
Inventory qty reconciliation	Opening qty + all IN - all OUT = closing qty, per branch/item/batch.
Inventory value reconciliation	Closing stock layers x historical cost = current inventory cost value.
COGS reconciliation	Stock sale-out cost - valid COGS reversals = product COGS.
Supplier reconciliation	Opening payable + credit purchases - payments - returns +/- adjustments = closing payable/advance.
Commission reconciliation	Earned - reversed - paid = commission outstanding/liability.
Receivable reconciliation	Effective invoice totals - net applied payments = outstanding receivable.
Profit reconciliation	Operating Profit uses recognized sales/cost/expense events, never bank balance changes as a proxy.

Automated regression tests to keep permanently

- Tax excluded from income, operating profit and staff commission under normal, discounted and refunded sales.
- Package commission example reconciles exactly after rounding and after partial component refund.
- Credit/cash/partial purchase produces correct stock, supplier payable and payment-account entries.
- Supplier payment changes payable/account only, never COGS.
- POS product sale uses historical batch cost; failed/retried posting cannot double deduct stock.
- In-house use reduces inventory and increases material-consumption cost with zero sales.
- Cross-date refund/tip/commission reversals remain in the reversal period without rewriting prior month.
- Payment Account opening + movement = closing for each account and branch.
- Operating Profit equals canonical component totals and excludes tax/tips/transfers.

15. Roles, Access and Implementation Sequence

15.1 Reporting access

Role	Reporting behavior
Super Admin / Owner	All authorized branches; consolidated read-only overview; filter any authorized user; explicit branch required for mutations outside reports.
Admin	Assigned branch reports and operational permissions only.
Accountant	Assigned-branch financial reports, POS/billing/dues and authorized supplier-payment/account functions; no silent

Role	Reporting behavior
	stock-admin privileges.
Staff	Personal attendance/performance/report access only; no general financial analytics unless explicitly granted.

15.2 Recommended build sequence

Phase	Deliverable
1. Current	Complete and verify Inventory + Suppliers + batch/expiry + stock movements + supplier ledger + POS product integration + COGS.
2. Reporting batch	Refactor/build the final 11 report pages using the formulas and data lineage in this document; remove extra reporting-menu items.
3. Reconciliation QA	Cross-report totals, date boundaries, refunds/voids, tax/tips, account and inventory reconciliation tests.
4. Frontend functional freeze	Close remaining portal/permission placeholders and dashboard source integrations.
5. Final global UI redesign	Apply the approved purple/magenta Behance-inspired theme, sliding navigation, card system and consistent responsive UX across all portals.
6. Backend/database	Node/Express/TypeScript/PostgreSQL integration and migration from mock contracts.
7. Final E2E	Data migration validation, permissions, idempotency, accounting reconciliation, browser QA and deployment preparation.

Implementation acceptance gate

Do not call the reporting batch complete until the same selected date/branch produces reconciling figures across Sales & Invoices, Payment Accounts, Inventory/COGS and Operating Profit, with tax/tips excluded from income and commission exactly as specified.

15.3 Final locked rules summary

- Tax is liability, never income, profit or commission base.
- Tips are staff liability, never income or commission base.
- Discounts reduce net sales before commission and tax basis calculations as configured.
- Commission uses staff-attributed net sales after discounts and reversals, excluding tax and tips.
- Inventory is valued at historical cost layers, not POS selling price.
- Purchases/supplier payments are not COGS. COGS occurs on product sale; salon material cost occurs on consumption.
- Bank/online accounts must always expose opening, money in, money out, transfers and closing balance.
- In-house inventory use is separately visible by quantity and cost.
- Original invoices, purchases and historical financial events are never deleted to make reports look current.
- Refunds, reversals and adjustments are dated events so historical periods remain correct.

END OF SPECIFICATION